Fiscal Year:	FY 2023	**Task Last Updated:**	FY 03/21/2023
PI Name:	Hayman née Anderson, Allison Ph.D.		
Project Title:	Multi-Environment Virtual Training for Long Duration Exploration Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80309-5004	**Congressional District:**	2
Comments:	NOTE: name change to Hayman née Anderson (Ed., March 2025). PI moved to University of Colorado from Dartmouth College in early 2017.		
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC019N0001-HFBP, OMNIBUS3 Crew Health: Human Factors and Behavioral Performance-Appendix E; Omnibus3-Appendix F
Start Date:	05/19/2021	**End Date:**	05/18/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	6	**No. of Master' Degrees:**	0
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clark, Torin Ph.D. (University of Colorado, Boulder) Robinson, Stephen Ph.D. (University of California, Davis) Vance, Eric Ph.D. (University of Colorado, Boulder) Szafir, Daniel (University of North Carolina)		
Grant/Contract No.:	80NSSC21K1140		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Historically, NASA has led the way in virtual reality (VR) technology development to train astronauts for microgravity environments, but there remain open questions regarding the use of adaptive VR for training in long duration exploration missions (LDEM). These missions are likely to have long time periods between pre-mission training and execution of a mission task. Crews will need to train and maintain skills on tasks critical to mission success, such as entry decent and landing (EDL), habitat maintenance and repair, and planetary surface extravehicular activity (EVA). To be successful, performance on these tasks will need to be maintained after high-fidelity training on Earth prior to the start of the mission, and refreshed during transit prior to execution on the surface. The objective of this research is to develop an effective training environment for mission critical LDEM tasks. We will experimentally investigate how skills transfer from the training environment to a physical setting, the degree to which the training environment enables skills maintenance over longer durations, and the generalizability of skills learned in the training environment when transferring to un-trained tasks. As a secondary benefit, we will investigate training in an adaptive, complex VR environment as a countermeasure to the neural decrements demonstrated in spaceflight. Thus, this research will quantify neurophysiological response in the cerebral cortex during training to assess the potential benefit of the trainer as a countermeasure. The novel contributions of this research are to investigate the use of immersive, adaptive VR environments to train astronauts specifically designed for LDEMs by 1) Incorporating multiple, mission critical tasks for a mission to Mars; 2) Investigating skills transfer, retention, and generalizability; 3) Designing with low volume, mass, and operational overhead; and 4) Investigating the degree to which the proposed training can also provide secondary benefits of broad brain stimulation. Our proposed research is directly aligned with that of the Virtual NASA Specialized Center of Research (VNSCOR), and can be integrated with researchers investigating teams, optimal adaptive control algorithms, motivational strategies, or neural and cognitive decrements with both human and animal studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has wide application in training in immersive environments, particularly for complex tasks. We develop a novel training algorithm to facilitate rapid skills acquisition, while also ensuring broad improvement in performance (rather than unidimensional performance). To our knowledge is absent from the literature, but may improve how we administer training in an automated (rather than supervised) training paradigm. This research will also contribute to our understanding of immersive environments and the degree to which they stimulate the brain. This informs our scientific understanding of neuroplasticity when learning and acquiring new skills. This has applications in rehabilitation, to inform how using these environments may yield better patient outcomes.
Task Progress:	Historically, NASA has led the way in virtual reality (VR) technology development to train astronauts for microgravity environments, but there remain open questions regarding the use of VR for training in long duration exploration (LDEM) missions. These missions are likely to have long time periods between pre-mission training and execution of a mission task. Crews will need to train and maintain skills on tasks critical to mission success, such as entry descent and landing (EDL), habitat maintenance and repair, and planetary surface extravehicular activity (sEVA). To be successful, performance on these tasks will need to be maintained after high-fidelity training on Earth pre-mission, and refreshed during transit prior to execution on the surface. The objective of this research is to develop an effective training environment for mission critical LDEM tasks. We will investigate how skills transfer from the VR training environment to a physical setting, the degree to which the training environment enables performance maintenance over longer durations, and the generalizability of skills learned in the training environment when transferring to un-trained tasks. As a secondary benefit, training in an adaptive, complex environment could serve as a countermeasure to the neural decrements demonstrated in spaceflight. Thus, this research will quantify neurophysiological response in the cerebral cortex during training to assess the potential benefit of the trainer as a countermeasure. In Year 2, we have made significant progress on developing an LDEM-specific, multi-environment virtual training simulator, called Trinity. Trinity incorporates three environments focused on Mars mission concepts: EDL, habitat maintenance/repair, and EVA using a surface rover. To date, we have completed development on the EDL task and have performed human subject experiments to assess skills transfer going from the virtual training environment to a high-fidelity test environment (physical mock-up). This training condition is compared to a 2D display and a non-adaptive VR training environment. We have also developed multiple adaptive training algorithms for complex tasks to achieve personalized acquisition of skills and have performed an experimental evaluation to assess the algorithm. Data analysis is ongoing. Development of the EVA task is nearing completion, and we are performing pilot testing to fine-tune the environment. The habitat maintenance and repair task is under development and a preliminary design review to finalize the task has been completed. The virtual reality environment and object interaction development has been completed. We will begin the evaluation of training as a countermeasure to neural decrements in spaceflight. For the corollary tasks, we have begun development recently. In the next year, we will quantify brain activation during training using functional near-infrared spectroscopy (fNIRS) and electroencephalogram (EEG). We have now hired an MS student who will perform this experimental evaluation in his thesis to be completed in Spring 2024. We have moved forward our Aim 3 to investigate skill retention using the EDL task. This experiment incorporates lessons learned from the Aim 1 experimental findings and is currently under Institutional Review Board (IRB) review. In future years, we will complete these ongoing investigations, focus on investigating VR training as a neurologic countermeasure, and demonstrate long-term acceptability, engagement, and skills transfer in an isolated, confined, controlled environment with other researchers in the Mars Adaptive TRaining-Integrative Knowledge System (MATRIKS) VNSCOR.
Bibliography Type:	Description: (Last Updated: 02/18/2026)
Abstracts for Journals and Proceedings	Verniani A, Galvin E, Tredinnick S, Vance E, Anderson A. "Evaluating astronaut training algorithms in virtual reality for long duration exploration missions. Poster." 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. , Feb-2023

Abstracts for Journals and Proceedings	Putman E, Boppana A, Peterson B, Clark T, Anderson A. "Immersive and adaptive training with virtual reality for long duration exploration missions." 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. Poster. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Anderson A, Putman E, Rees W, Peterson B, McVey C, Verniani A, Schlittenhart M, Mayali M, Szafir D, Clark T. "Trinity: Multi-environment virtual trainer for long duration exploration missions." 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. Poster. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Verniani A, Putman E, Boppana A, Peterson B, Tredinnick S, Galvin E, Vance E, Clark T, Anderson A. "Efficacy of adaptive algorithms for training in virtual reality." 2023 Annual Scientific Meeting, Aerospace Medicine Association (AsMA), New Orleans, Louisiana, May 21-25, 2023. Accepted Presentation. Abstracts. 2023 Annual Scientific Meeting, Aerospace Medicine Association (AsMA), New Orleans, Louisiana, May 21-25, 2023. , May-2023
Abstracts for Journals and Proceedings	Putman EJ, Boppana A, Clark TK, Anderson AP. "Adaptive training using virtual reality for entry, descent, and landing during long duration exploration missions." IAC 2022: 73rd International Astronautical Congress, Paris, France, September 18-22, 2022. Abstracts. IAC 2022: 73rd International Astronautical Congress, Paris, September 18-22, 2022. IAC Paper ID 72408. https://dl.iafastro.directory/event/IAC-2022/paper/72408/ , Sep-2022
Awards	Anderson A. "Outstanding Research Mentor - University of Colorado Boulder "Summer Program for Undergraduate Research" (CU SPUR) Program, August 2022." Aug-2022